

## DBMS Short note

### Week 1

- Data – the Data is and kind of Fact and Figures can be saved somewhere
- Information – information is data that is meaningful and processed

#### Data vs Information

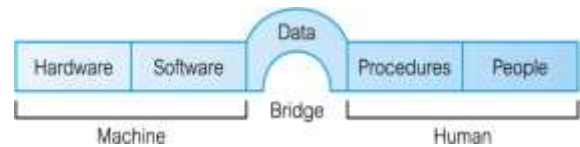
| DATA                   | INFORMATION          |
|------------------------|----------------------|
| Row facts              | Final result         |
| No context             | With context         |
| Useless                | Useable              |
| Before processed       | Processed data       |
| Just number and text   | Just number and text |
| etc... without meaning | etc... with meaning  |

#### File based system

1. Manual file Processing (paper based)  
Drawbacks
    - Time Consuming
    - Does not support large volumes of data
  2. File based Processing  
Drawbacks
    - Data duplication
    - Inconsistent Data
    - Inflexibility
    - Limited data sharing
    - Excessive program maintenance
  3. Database Management system  
Advantage
    - Minimal data redundancy
    - Data integrity and security
    - Restricting Unauthorized Access
    - Minimized data inconsistency  
Drawbacks
    - Complexity
    - Additional Hardware costs
    - Experts – Specialised personnel
    - Higher impact of failure
- Database - A shared collection of logically related data designed to meet the information requirements of an organisation.

- DBMS - DBMS is a collection of programs that manages the database structure and controls access to the data stored in the database

#### Components of DBMS Environment



#### When not to use a DBMS

- High initial investment in hardware, software, and training.
- Overhead for providing security, concurrency control, recovery, and integrity functions.

#### Main Characteristics of the Database Approach

1. Self-describing nature of a database system
  2. Insulation between programs and data
  3. Data Abstraction
  4. Support of multiple views of the data
- Data Abstraction -The characteristic that allows program-data independence and program-operation independence is called data abstraction.
- data model - Data models define how data is connected to each other and how they are processed and stored inside the system

week 2-3

- Database architecture - The database architecture is the set of specifications, rules, and processes that dictate how data is stored in a database and how data is accessed by components of a system

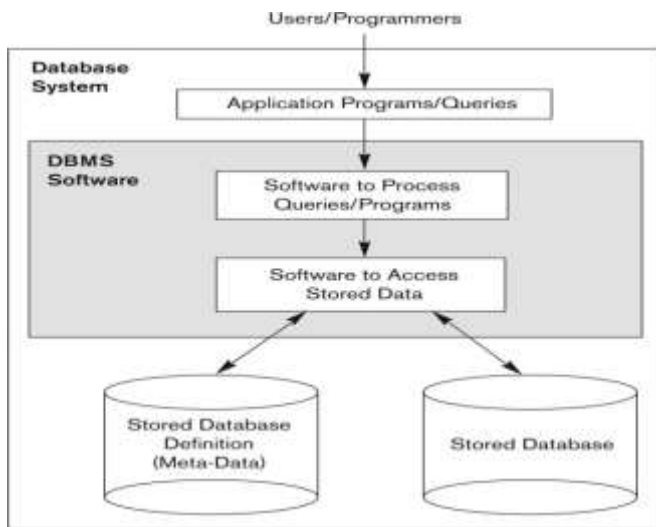
#### Basic Client-Server Architectures

- Clients
- DBMS Server

#### Database Architecture

1. **Logical 2 tier client server architecture**  
Is used for user interface programs and application programs that run on client side.
2. **Logical 3 tier client server architecture**  
This is used in web application

## A simplified database system environment



### What do we store in a Database?

- Collection of data central to some enterprise
- Essential to operation of enterprise
- Historical data can guide enterprise strategy
- State of database mirrors state of enterprise

➤ **Data Model:** A set of concepts to describe the structure of a database and certain constraints that the database should obey

### Categories of data models

- Conceptual (high-level, semantic)
- Physical (low-level, internal)
- Implementation (representational)
- 

#### Database model (implementation)

- **Hierarchical**
  - **Network**
  - **Relational** - Current Generation
  - **Object Based** - Advanced Systems
- Early Types of DBMS

#### Hierarchical Model

##### Advantages

- Conceptual simplicity
- Database security
- Data independence
- Database integrity

##### Disadvantages

- Complex implementation

- Difficult to manage for complex sets of data
- Lacks structural independence

## Network Model

### Advantages

- Conceptual simplicity
- More relations can be handled
- Data independence
- Database integrity

### Disadvantages

- Complexity increases with large databases and multiple relation types
- Difficult to make structural changes
- Database design and update activities require more time

## Relational Model

### Advantages

- Dynamic links among tables
- Easy and flexible to design and use
- Data independence
- Generic manipulation language (SQL)

➤ **Schema:-** A description of data in terms of a data model is called the **schema**.

### Schemas versus Instances

- Database Schema
- Schema Diagram
- Schema Construct
- Database Instance

### Database Schema Vs. Database State

**Database schema :** changes *very infrequently*

**Database state :** changes *every time the database is updated*.

### Database Schema

A database schema is the skeleton structure that represents the logical view of the entire database.

### Three-Level Schema Architecture

#### 1. Physical/internal schema

The internal level describes the physical storage structure of the database

## 2. Conceptual/logical schema

The conceptual level describes the structure of the whole database for a community of users.

## 3. External schemas

Allow data access to be customized (and authorized) at the level of individual users or groups of users

### Proposed to support DBMS characteristics of

- Program-data independence.
- Support of multiple views of the data.

### Data Independence

Data independence helps you to keep data separated from all programs that make use of it

- **Logical Data Independence:** The capacity to change the conceptual schema without having to change the external schemas and their application programs.
- **Physical Data Independence:** The capacity to change the internal schema without having to change the conceptual schema.

### Relational Objects

- **Primary Key:** An attribute that uniquely identifies each row in a relation.
- **Composite Key:** A primary key that consists of more than one attribute
- **Cardinality:** The cardinality of a table refers to the number of rows in the table
- **Degree:** The degree of a table refers to the number of columns.
- **Super Key** – An attribute or a combination of attribute that is used to identify the records uniquely is known as Super Key.
- **Candidate Key** – Candidate key is a single key or a group of multiple keys that uniquely identify rows in a table
- **Entity integrity** - Entity integrity constraint specifies that no primary key can be null
- **foreign keys-( referential integrity)**- A foreign key is a set of columns in one table that serve as the primary key in another table
- **Recursive foreign key:** A foreign key in a relation that references the primary key values of that same relation

- **Query** - A query is a question about the data in your tables.

### DBMS Languages

- **Data Definition Language(DDL)**
  - Used to create and modify database objects  
Create, Drop, Alter  
**storage definition language (SDL)**  
**view definition language (VDL)**
- **Data Manipulation language(DML)**
  - Used to create, modify and retrieve data  
Insert, Select, Update, Delete
- **Data Control Language(DCL)**
  - is a syntax similar to a computer programming language used to control access to data stored in a database.

### Transaction Control Language (TCL)

- These SQL commands are used for managing changes affecting the data.  
COMMIT, ROLLBACK, SAVEPOINT.

### SQL - Structured Query Language

- *Used as DML,DDL and DCL (TCL)*

### Types And Classification Of Database Management System

#### 1. Based on the data model

Hierarchical, Network, Relational, Object-oriented

#### 2. Based on the number of users

- Single user
- Multiple users

#### 3. Based on the sites over which network is distributed

- Centralized database system
- Parallel network database system
- Distributed database system

#### 4. Based on the usage

- Online transaction processing(OLTP) DBMS
- Online analytical processing(OLAP) DBMS

### **Database Design**

- Database design process can be divided into 6 major steps:
  1. Requirements Analysis
  2. Conceptual Database Design
  3. Logical Database Design
  4. Schema Refinement
  5. Physical Database Design
  6. Security Design